

AI in mixed reality – Copilot on HoloLens

Spatial computing with large language models

Andy Klein

Principal Mixed Reality Designer, Microsoft, Canada
anklei@microsoft.com

Ethan Arnowitz

Senior Designer, Microsoft, Canada
etarnowi@microsoft.com



Figure 1: Multi-factor intention determination provides the user with a taskbar attached to their hand. This taskbar attachment only occurs when they raise their palm (factor 1, skeletal hand pose detection) and look at their palm (factor 2, eye tracking raycast intersects the raised palm). This interface system is useful in avoiding accidental activations when working with tools and parts held in hands.

ABSTRACT

Mixed reality together with AI presents a human-first interface that promises to transform operations. Copilot can assist industrial workers in real-time with speech and holograms; generative AI is used to search technical documentation, service records, training content, and other sources. Copilot then summarizes to provide interactive guidance.

CCS CONCEPTS

• **Human-centered computing**; • **Information systems**; • **Information retrieval**; • **Specialized information retrieval**; • **Environment-specific retrieval**; • **Computing methodologies**; • **Computer graphics**; • **Graphics systems and interfaces**; • **Mixed / augmented reality**;

KEYWORDS

3D, AI, AR, Mixed Reality, Spatial Computing, HoloLens, LLM, Gestures, Voice, User Experience Design, Real-time

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1 INTRODUCTION

Using computer vision, speech to text, text to speech, gesture recognition, LLMs, and curated 2D and 3D datasets to identify and describe technical information at time of user request, for on the job performance support. This technology is empowering industrial technicians facing overwhelming complexity in aerospace, automotive, energy, pharmaceutical, and semiconductor manufacturing scenarios.

Our system leverages the user’s spoken phrases and operates on a closed data universe to retrieve only facts from the curated dataset. It orchestrates input from HoloLens runtimes for eye tracking and hand tracking in 3D space to discover valuable information from the scene for additional context about the user and the objects around them. The resulting generative AI text phrases playback as audio and print in the chat history log. The generated phrases are coupled with relevant visualizations of 3D parts overlaid as spatial holograms on the machinery. The user can move around their space and the holograms remain attached to the machine as their work progresses.

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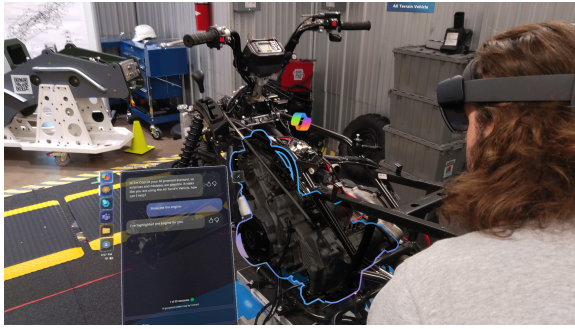


Figure 2: A user interacting with Copilot sees 3D data overlays in mixed reality. Their AI conversation window contains a historic record of their interactions for reference, including citations to source documents used in answer generation.

While curation of the dataset is an important piece of the customer workflow, this presentation focuses on consumption of curated data at runtime, to provide real-time procedural support to users. Dynamics 365 Dataverse tables host and store customer data, in the form of technical documents and 3D hierarchies. This information is indexed and chunked for retrieval through a real-time service. Specific LLM agent prompting instructs Copilot to only respond with information found in the customer's uploaded data. The HoloLens runtime loads a specific asset which links the data to the device before the user query session begins.

Leveraging 3D scene graphs with gesture or gaze based raycasts and collisions, coupled with linguistic generative AI interpretation of speech, our mixed reality Copilot provides contextual audio-visual responses in a spatialized experience on HoloLens. The system relies on an anchored hierarchical 3D dataset overlaid on the real world.

This presentation showcases what a work-focused spatial computing device can do to provide value to an audience of frontline technicians who is historically underserved by technology. We will demonstrate the latest functions for using mixed reality computers to achieve repeatable high quality work outcomes, even when the user doesn't have a desk or their hands are too busy to use a phone in order to benefit from digital information. Without a desk, these users can't have a laptop or desktop computer. With perpetually busy hands, these users can't use a phone. HMDs like HoloLens are purpose-built for these situations, to help people who need to benefit from digital information in the course of their complex work.

- Use cases and user demographics span training and industrial manufacturing scenarios.
- A spatial taskbar is always available at the user's fingertips but never in the way.
- Conversational AI provides spatial outcomes via user speech and curated data.
- Gestural input to 3D scene provides additional context for LLM processing.
- 3D object detection identifies and anchors information to the real world.



Figure 3: When working in complex environments and procedures, users can point and ask about any part of a machine to get real time AI support without needing to know all of the specific component names. The AI orchestration runtime can selectively highlight relevant elements from the anchored 3D scene hierarchy based on the context of the user's query and conversation history.

We have built a workflow which enables fast data curation and fast data extraction at time of need. This system meets the requirements for deployment of technology into workstreams and workflows which previously relied on paper manuals or other manual systems. These new workflows enable customers to capture all of the wisdom from their datasets and make it available to users on HoloLens for performing tasks in the physical environment away from screens and traditional computers.